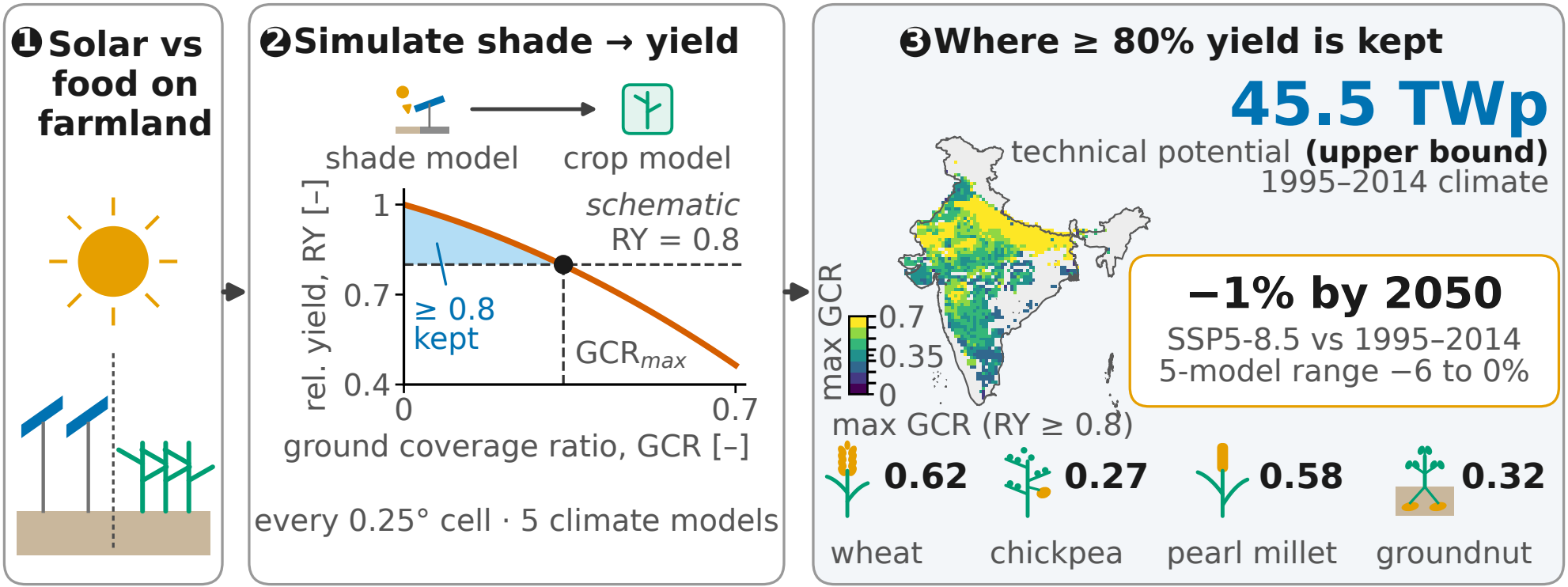

❶ Solar vs food on farmland
❷ Simulate shade → yield
shade model
crop model
rel. yield, RY [–]
1
0.7
0.4
0
0.7
schematic
RY = 0.8
≥ 0.8 kept
GCR$_{max}$
ground coverage ratio, GCR [–]
every 0.25° cell · 5 climate models
❸ Where ≥ 80% yield is kept
45.5 TWp
technical potential (upper bound)
1995–2014 climate
max GCR
0.7
0.35
0
max GCR (RY ≥ 0.8)
−1% by 2050
SSP5-8.5 vs 1995–2014
5-model range −6 to 0%
0.62
0.27
0.58
0.32
wheat
chickpea
pearl millet
groundnut